Aliah University
End Semester Examination 2020

AI 131: Elementary Arabic & Islamic Studies

Full Mark: 40   Time: 2:00 Hrs.

Group-A   20 Marks
Arabic

Attempt any Five Questions. Each question carries equal marks

1. Write down all the Arabic alphabets.
2. join the following letters and make complete words:

   م+س+ل+م=   ش+ا+هـ+د=   ق+و+ل=   ل+ا+ع+ب=   م+د+ر+س=

3. Use definite article before the below mentioned words:

   عالم، كتاب، شاعر ، سامع، معلمة

4. Write down all the Sun letters .
5. change the gender of the following words:

   عاقل، معلمة، سارق، كاتبة، عالم

6. Give plural forms of the followings:

   ذاهب، شاهد، شاعر ، صادق، ساطع

7. Discuss the importance of Arabic language briefly.

Group-B   20 Marks
Islamic Studies

Attempt any Four Questions. Each question carries equal marks

1. Discuss the religious condition of the Arabs before the arrival of Islam.
2. write a short note about the five pillars of Islam.
3. what do you know about early life of Prophet Muhammed (PBUH)?
4. Give a brief account of the five pillars of Islam.
5. write a short note about the battle of Badar.
6. what is the Treaty of Hudaybia? Discuss briefly.

Email for Submitting the Answer Scrip: guestfaculty1.islamicstudies@aliah.ac.in

ALIAH UNIVERSITY

B.Tech End Semester (Odd) Examinations 2021
Subject– Basic Electronics Engineering (ESC ECEUGES01)
CSE , CEN, MEN 1st Year 1st SEM

Full Marks : 80 **Time-3 hrs**

(Answer any five)

1. (i) Briefly explain the formation of energy band in solid state materials.
(ii) Distinguish between insulator , semiconductor and metal on the basis of energy band diagram.
(iii)Differentiate between intrinsic and extrinsic semiconductor. (6+6+4)

2. (i) How is an n-type semiconductor formed ?
(ii) Explain the Fermi-Dirac distribution function with graphical interpretation.
(iii) Show that in n-type semiconductor the Fermi level energy E_F is close to the conduction band edge E_C of the band diagram. (3+8+5)

3. (i) Show that equilibrium concentrations of electrons (n_0) and holes (p_0) in a semiconductor can be expressed as :- (symbols have usual meaning)

(a) $n_0 = N_c e^{-\left(\frac{E_C - E_F}{KT}\right)}$ (b) $p_0 = N_v e^{-\left(\frac{E_F - E_V}{KT}\right)}$

(ii) Show that the product of electron and hole concentrations under equilibrium is constant and can be expressed as :- $n_0 p_0 = n_i^2$. (symbols have usual meaning) (6+6+4)

4. (i) Explain the formation of the depletion region and barrier potential V_0 in a p-n junction diode.
(ii) Show that the barrier potential can be expressed as :- (symbols have usual meaning)

$$V_0 = \frac{KT}{q} \ln \frac{N_a N_d}{n_i^2}$$ (4+4+8)

5. (i) Explain the operation of the clipper circuit with the input shown in **Fig. 1**, assuming the diode as ideal diode.
(ii) Find out the changes in the output voltage waveform if the diode is a real one with cutin voltage v_γ and diode dynamic resistance r_d after estimating the output voltage expression.
(iii) Draw and explain the output of the circuit shown in **Fig. 2** for the given input. (5+5+6)

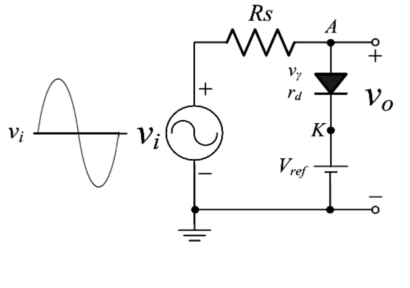

Fig. 1

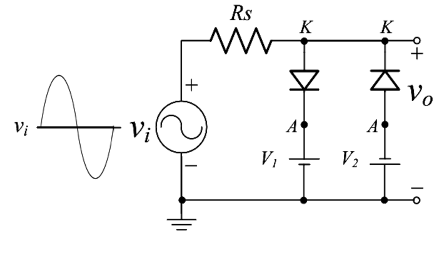

Fig. 2

6. (i) Explain the operation of full wave rectifier circuit with bridge network with a output waveform diagram.
(ii) Show that the output DC voltage (V_{dc}) and ripple factor (Γ) of full wave rectifier output is

(a) $V_{dc} = 0.636\, V_m$ (b) $\Gamma = 48.3\ \%$

(4+5+3+4)

7. (i) Explain different current components in a p-n-p transistor in active region with a suitable diagram.

(ii) Establish the relation between current amplification factors α and β. (6+4+6)

(iii) Briefly explain the CB, CE and CC mode operation in a transistor.

8. (i) Draw the output characteristics of CE amplifier and explain it.

(ii) With neat diagram explain the fixed bias circuit of CE amplifier.

(iii) Draw the DC load line and Q-points of the CE amplifier circuit. Why is the Q-point need to remain in the middle of the load line? (4+4+6+2)

9. Find the input-output relation of the circuits shown in Fig. 3 and Fig. 4. (8+8)

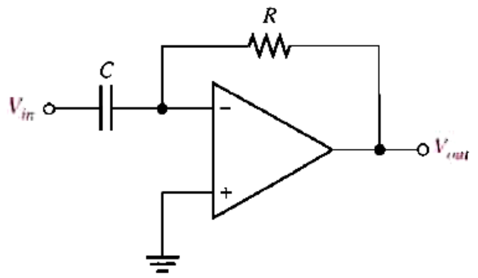

Fig. 3

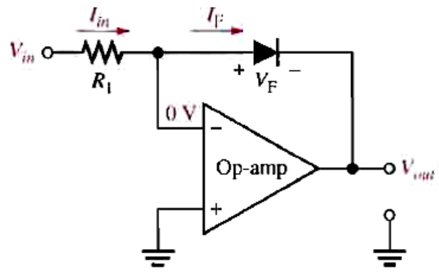

Fig. 4

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ALIAH UNIVERSITY
DEPARTMENT OF MATHEMATICS AND STATISTICS
Odd Semester End Examination, 2021
COURSE: ENGINEERING MATHEMATICS COURSE CODE: MATUGBS01
Total marks : 8x10=80 **Time allowed : 3 hours**

Candidates are requested to read the following instructions carefully before answering the questions:

Instructions: Answer any **EIGHT** questions. The marks in each questions are equally distributed.

1. (a) Find the values of a and b such that
$$\lim_{x\to 0} \frac{x(1+a\cos x)-b\sin x}{x^3}=1.$$
(b) Find the value of k, for which given function
$$f(x)=\begin{cases}\left\{\tan(\frac{\pi}{4}+x)\right\}^{\frac{1}{x}}, & \text{when } x\neq 0,\\ k, & \text{when } x=0.\end{cases}$$
is continuous at $x=0$.

2. (a) If $y=a\cos(\log x)+b\sin(\log x)$, show that
$$x^2y_{n+2}+(2n+1)xy_{n+1}+(n^2+1)y_n=0.$$
(b) If
$$V=\log(x^3+y^3+z^3-3xyz),$$
prove that
$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 V=-\frac{9}{(x+y+z)^2}.$$

3. (a) State Euler's theorem for homogeneous functions. If $u=\log\left(\frac{x^4+y^4}{x+y}\right)$, show by Euler's theorem that
$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=3.$$
(b) Let
$$f(x,y)=\begin{cases}\frac{xy(x^2-y^2)}{x^2+y^2}, & \text{when } (x,y)\neq(0,0),\\ 0, & \text{when } (x,y)=(0,0).\end{cases}$$
show that $f_{xy}(0,0)\neq f_{yx}(0,0)$.

4. (a) Starting from $\frac{dx}{dy}=\frac{1}{\frac{dy}{dx}}$, obtain
$$\frac{d^2x}{dy^2}=-\frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3} \text{ and } \frac{d^3x}{dy^3}=-\frac{\frac{d^3y}{dx^3}\frac{dy}{dx}-3\left(\frac{d^2y}{dx^2}\right)^2}{\left(\frac{dy}{dx}\right)^5}.$$

(b) Find the condition for the line $x \cos\theta + y \sin\theta = p$ to touch the curve $\frac{x^m}{a^m} + \frac{y^m}{b^m} = 1$.

5. (a) Find the Fourier cosine series of $f(x) = x^2$ on the interval $[-\pi, \pi]$.

   (b) Find the Fourier series of the function

$$f(x) = \begin{cases} 0, & -5 < x < 0, \\ 3, & 0 < x < 5. \end{cases}$$

6. (a) Evaluate:

$$\int \frac{\sin x \cos x}{a^2 \cos^2 x + b^2 \sin^2 x}\, dx.$$

   (b) Show that

$$\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x + \cos x} dx = \frac{1}{\sqrt{2}} \log(1 + \sqrt{2}).$$

7. (a) Evaluate the limit:

$$\lim_{n\to\infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + ... + \frac{1}{n+3n} \right]$$

   as an integral.

   (b) Examine the convergence of

      i.
$$\int_0^2 \frac{dx}{(2x - x^2)},$$

      ii.
$$\int_{-\infty}^{\infty} \frac{dx}{1 + x^2}.$$

8. (a) Prove that

$$B(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1 + x)^{m+n}} dx,$$

   where B denotes the Beta function.

   (b) Write the relation between Beta and Gamma functions. Using the definition of Beta function find the value of the integral:

$$\int_0^{\frac{\pi}{2}} \sin^5\theta \cos^7\theta d\theta.$$

9. (a) State the Cauchy Mean value theorem. If in the Cauchy Mean value theorem $f(x) = e^x$ and $g(x) = e^{-x}$, show that c is arithmetic mean between a and b,

   (b) Evaluate:

$$\lim_{x\to 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$$

.

10. (a) If $u = \tan^{-1}\left(\frac{x^{\frac{5}{7}}+y^{\frac{5}{7}}}{\sqrt{x}-\sqrt{y}}\right)$ show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$.

(b) If

$$v = z\sin\frac{y}{x} \quad \text{where} \quad x = 3r^2 + 2s,\ y = 4r - 2s,\ z = 2r^2 - 3s^2$$

then find $\frac{\partial v}{\partial r}$ and $\frac{\partial v}{\partial s}$.

11. (a) Check whether the limits exist or not at the origin, if exist then find limit(/s) for the functions given below:

i. $f(x,y) = \begin{cases} \frac{x^4+y^4}{x-y}, & \text{when } x \neq y, \\ 0, & \text{otherwise.} \end{cases}$

ii. $f(x,y) = \begin{cases} \frac{x^3 y}{x^4+y^2}, & \text{if } (x,y) \neq (0,0), \\ 37, & \text{if } (x,y) = (0,0). \end{cases}$

.

(b) If $ax + by = 1$ is normal to the parabola $y^2 = 4\lambda x$, then prove that $\lambda a^3 + 2a\lambda b^2 = b^2$.

12. (a) State Lagrange's Mean value theorem. If $f(x) = (x-1)(x-2)(x-3)$, verify Lagrange's theorem in the interval $[0,4]$.

(b) If $y = \cos(m\sin^{-1} x)$, then prove that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0.$$

Hence prove that

$$\begin{aligned} (y_n)_0 &= 0, \ \textit{if } n \textit{ be odd}, \\ &= -m^2(2^2 - m^2)(4^2 - m^2)\ldots\{(n-2)^2 - m^2\}, \ \textit{if } n \textit{ be even.} \end{aligned}$$

Autumn-Semester Examination- 2020

Course code: PHYUGBS-01
Course title: Engineering Physics
Department of Appearing Students: CEN, CSE, ECE, EEN, MEN
Full marks: 80 **Time: 3 hrs**

Group-A

(Answer any four questions from below.)

$(10 \times 4 = 40)$

1. (a) Write down the first law of thermodynamics. (b) If the degree of freedom of a gas is n, then find the ratio of C_P and C_V . (c) State and prove the law of equipartition of enengy. (2+3+5)

2. (a) What do you mean by the 'thermodynamic equilibrium', 'macroscopic parameter'? (b) An ideal gas expands adiabatically but quasistically from the state $i(P_i, V_i, T_i)$ to the state $f(P_f, V_f, T_f)$. What is the change in its internal energy ? What would this change be if the expansion were isothermal? (3+7)

3. (a) Write down the second law of thermodynamics (Kelvin-Plank Statement). (b) Find out the entropy of the steam. (c) If the equation of state for a gas with internal energy U is $PV = \frac{U}{3}$, then find out the equation for an adiabatic process. (2+4+4)

4. (a)What is refrigerator? Find out the co-efficient of performance of a refrigerator. (b) Discuss the four basis postulates of the kinetic theory of gases. (6+4)

5. Prove the thermodynamics relations;
   (a)$TdS = C_V dT + T\left(\frac{\partial P}{\partial T}\right)_V dV$
   (b) $TdS = C_P dT - T\left(\frac{\partial V}{\partial T}\right)_P dP$
   (c) For homogeneous fluid, $C_P - C_V = T\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_P$ (3+3+4)

Group-B

(Answer any four questions from below.)

$(10 \times 4 = 40)$

6. (a) Define unit vector and null vector. (b) Show that $\vec{\nabla} \times \vec{F} = 0$, where $\vec{F} = (2xy + z^3)\hat{\imath} + x^2y\hat{\jmath} + 3xz^2\hat{k}$. (c) Find out $\vec{\nabla} \cdot \vec{r}$, where r is the position vector. (d) Find out $\vec{A} \cdot (\vec{B} \times \vec{C})$,where $\vec{A} = 2\hat{\imath} - 5\hat{\jmath}$, $\vec{B} = 3\hat{\jmath}$, $\vec{C} = 5\hat{\jmath} + 6\hat{k}$. (e) Can the scalar product of two vectors be negative? (2+2+2+2+2)

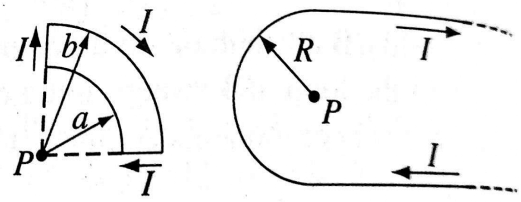
7. (a) If $\vec{A}$ and $\vec{B}$ are two length vectors, then what is the geometrical significance of $|\vec{A} \times \vec{B}|$? (b) Find a vector which is perpendicular to each vectors: $\vec{A} = 2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}$ and $\vec{B} = \hat{\imath} + \hat{\jmath} - \hat{k}$. (c) Calculate the area of the parallelogram where two adjacent sides are formed by the vectors $\vec{A} = 3\hat{\imath} + 4\hat{\jmath}$ and $\vec{B} = -3\hat{\imath} + 7\hat{\jmath}$. (d) Find the condition for two vectors to be parallel and perpendicular to each other. (e) Find the workdone in moving a paricle along a vector $\vec{S} = (4\hat{\imath} - \hat{\jmath} + 7\hat{k})$ if the applied force is $\vec{F} = (\hat{\imath} + 2\hat{\jmath} - \hat{k})$ newton, $\vec{S}$ is meter. (2+2+2+2+2)

8. (a) Define electric field at a point? (b) Write down the unit and dimension of electric field? (c) Find the electric field at a distance z above the centre of a flat circular disk of radius $\vec{R}$ that carries a uniform surface charge σ. What does your formula give in the limit $R \to \infty$? Also check the case $z \gg R$. (d) Wriite down the diffenrential form of Gauss's law? (2+2+4+2)

9. (a)Define lattice, Basis and Crystral structure? (b) What is the maximum number of possible Bravis lattice in 3D? (c) Derive the expression of interplanar spacing in 3D. (d) Find the lattice indices of a plane that makes intercepts of 1 on a-axis, 2 on b-axis and is parallel to c-axis. (e) What is the number of effective atom present in a Diamond structure in a unit cell? Draw (101) and (111) planes in a cubic unit cell. (3+1+2+2+2)

10. (a) Write down Biot-Savart law? Express it in vector form. (b) A current is set up in a long copper pipe. Is there a magnetic field inside and outside the pipe? (c) Find the magnetic field at point P for each of the steady current configurations shown in fig(a) and fig(b).

(4+4+2)



(a) (b)

Autumn-Semester Examination- 2020

Course code: PHYUGBS-02
Course title: Engineering Physics Lab
Full marks: 70 Time-3 hrs

(A) Answer the following questions. (2x25=50)

1. Write the method to determine the thickness of a glass plate by Screw Gauge .(25)
2. Write the method to determine the dimension of the object by Vernier Caliper.(25)

(B) Write down the answer of following questions. (2x10=20)

1. What is the least count of Vernier Caliper?
2. What is least count of Screw Gauge?
3. State ohm's law .
4. Why we use a Vernier Caliper?
5. Why we use a Screw gauge?
6. What is Hooke's law.?
7. What are the use of multimeter?
8. How to make OR gate using NAND gate?
9. What is the resistance of an ideal ammeter?
10. What is the resistance of an ideal voltmeter?

“UG” END SEMESTER EXAMINATION – MARCH, 2021
(REGULAR)
1^{ST} YEAR B.TECH. MECHANICAL ENGINEERING
ENGINEERING MECHANICS
MENUGES01

[Full Marks : 80] **[Time : 3:00 hrs.]**

Assume suitable data(s), if not provided

A. Fill in the blanks/Multiple choice questions (*Compulsory*) – $[\mathbf{10X1 = 10}]$

1. The magnitude of the moment about the base point O of the $600\ N$ force (figure A1) is –

……………………

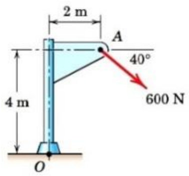


Figure A1

2. Which of the following are vector quantity(ies)?
a) Linear Displacement b) Linear Velocity
c) Linear Acceleration d) All of these

3. The point, through which the whole weight of the body acts, irrespective of its position, is known as –
a) center of mass b) centroid c) moment of inertia d) center of gravity

4. If the resultant of two forces P and Q acting at an angle θ makes an angle α with P, then the resultant should be – ……………………….

5. If two forces each equal to P in magnitude act at right angles, their effect may be neutralized by a third force acting along their bisector in opposite direction whose magnitude is –
…………………………

6. A framed structure is perfect if it contains members equal to –
a) $2n-3$ b) $3n-2$ c) $2n-1$ d) $n-2$

7. In the figure A2, angle α compared to β will be –
a) Greater b) Lower
c) Same d) Unpredictable

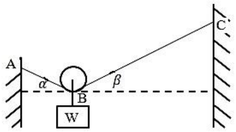


Figure A

8. Forces $5i + 6j + 7k\ N$ and $-8i - 9k\ N$ are acting at a point. The resultant force will be of magnitude –
……………………

9. The three forces keep acting at a point are in equilibrium, then –
a) The forces must be coplanar b) The forces need not be coplanar
c) The forces must be of equal magnitude d) The forces must be of unequal magnitude.

10. A mass of 10 kg is resting on a rough table. The friction force acting on it is –
………………………

B. Short questions (*Answer any five questions*): $[\mathbf{5X6 = 30}]$

1. The forces $\boldsymbol{F_1}, \boldsymbol{F_2}$ *and* $\boldsymbol{F_3}$, all of which act on point **A** of the bracket, as shown in the figure B1. Determine the resultant force and its direction.

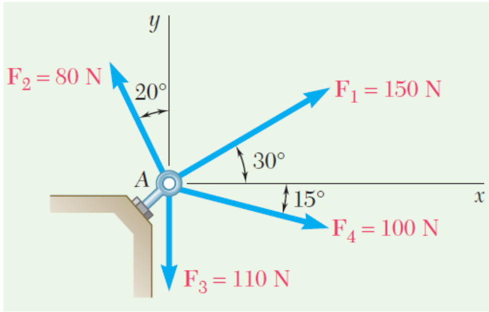


Figure B1

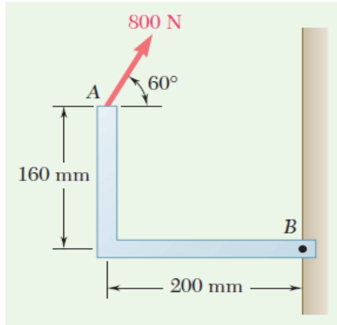


Figure B2

2. A force of 800 N acts on a bracket as shown in figure B2. Determine the moment of the force about B.

3. Locate the centroid of a circular sector with respect to its vertex. (figure B3).

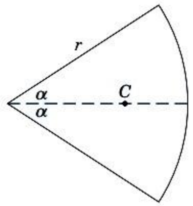


Figure B3

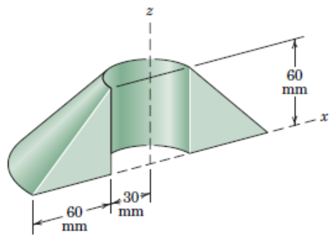


Figure B4

4. Calculate the volume V of the solid generated by revolving the 60-mm right-triangular area through 180^0 about z-axis. If this body were constructed of steel, what would be its mass “m”? (figure B4).

5. The position coordinate of a particle which is confined to move along a straight line is given by $s = 2t^3 - 25t + 7$, where s is measured in meters from a convenient origin and t is in seconds. Determine –

a) The time required for the particle to reach a velocity of $72\, m/s$ from its initial condition at $t = 0$,
b) The acceleration of the particle when $v = 30\, m/s$,
c) The net displacement of the particle during the interval from $t = 1s$ to $t = 4s$

6. A 75-kg man stands on a spring scale in an elevator. During the first 3-seconds of motion from reset, the tension “T” in the hoisting cable is 8300N. Find the reading R of the scale in newton during this interval and the upward velocity “v” of the elevator at the end of the 3-seconds. The total mass of the elevator, man and scale is 800 kg.

C. Long questions (*Answer any four questions*): $[4X10 = 40]$

1. For the vectors $\boldsymbol{V_1}$ and $\boldsymbol{V_2}$ shown in the figure C1,
- **a)** Determine the magnitude S of their vector sum $\boldsymbol{S} = V_1 + V_2$
- **b)** Determine the angle $\boldsymbol{\alpha}$ between $\boldsymbol{S}$ and the positive x-axis.
- **c)** Write $\boldsymbol{S}$ as a vector in terms of the unit vectors $\boldsymbol{i}$ and $\boldsymbol{j}$ and then write a unit vector $\boldsymbol{n}$ along the vector sum $\boldsymbol{S}$.
- **d)** Determine the vector difference $\boldsymbol{D} = V_1 - V_2$

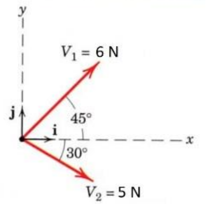
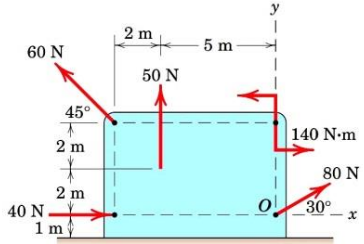


Figure C1

2. Determine the resultant of the four forces and one couple which act on the plate shown in figure C3.



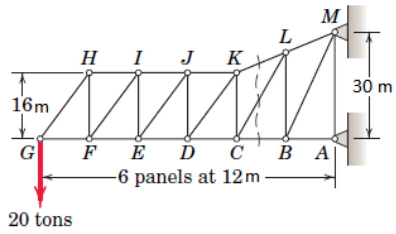


Figure C3 **Figure C4**

3. Compute the forces induced in members KL, CL and CB by 20 ton load on the cantilever truss (figure C4).

4. Determine the range of values which the mass m_0 may have so that the 100 kg block shown in the figure C5 will neither start moving up the plane nor slip down the plane. The coefficient of static friction for the contact surface is 0.20. (figure C5)

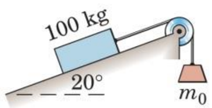


Figure C5

5. Each of the two uniform hinged bars has a mass m and length l, and is supported and loaded as shown. For a given force P, determine the angle θ for equilibrium. (figure C6)

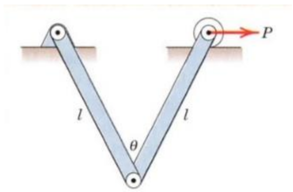


Figure C6

6. A tennis player strikes the tennis ball with her racket when the ball is at the uppermost point of its trajectory as shown. The horizontal velocity of the ball just before impact with the racket is $v_1 = 50\ m/s$, and just after impact its velocity is $v_1 = 70\ m/s$, as directed at the 15^0 angle as shown. If the 20-gm ball is in contact with the racket for 0.02 sec, determine the magnitude of the average force **R** exerted by the racket on the ball. Also determine the angle β made by **R** with the horizontal. (figure C7)

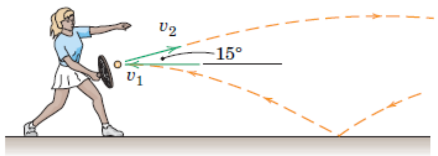


Figure C7

====== END ======